전산화학 (2011년 1학기) 2차 과제 해답

(총 40 점)

1. **(총 20점)**

$$x = r\cos\phi, \quad y = r\sin\phi \Rightarrow r^2 = x^2 + y^2, \quad \tan\phi = \frac{y}{x}. \tag{1}$$

연쇄법칙을 이용하면,

$$\left(\frac{\partial}{\partial x}\right)_y = \left(\frac{\partial r}{\partial x}\right)_y \left(\frac{\partial}{\partial r}\right)_\phi + \left(\frac{\partial \phi}{\partial x}\right)_y \left(\frac{\partial}{\partial \phi}\right)_r, \tag{2a}$$

$$\left(\frac{\partial}{\partial y}\right)_x = \left(\frac{\partial r}{\partial y}\right)_x \left(\frac{\partial}{\partial r}\right)_\phi + \left(\frac{\partial \phi}{\partial y}\right)_x \left(\frac{\partial}{\partial \phi}\right)_r, \tag{2b}$$

이고,

$$\frac{\partial r}{\partial x} = \frac{x}{r} = \cos\phi, \quad \frac{\partial \phi}{\partial x} = -\frac{\sin\phi}{r}, \tag{3a}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r} = \sin\phi, \quad \frac{\partial \phi}{\partial y} = \frac{\cos\phi}{r}, \tag{3b}$$

이므로,

$$\boxed{\begin{aligned} \frac{\partial}{\partial x} &= \cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial \phi}, \quad (4a) \\ \frac{\partial}{\partial y} &= \sin\phi\frac{\partial}{\partial r} + \frac{\cos\phi}{r}\frac{\partial}{\partial \phi}. \quad (4b) \end{aligned}}$$

한편, g 를 다음과 같이 정의하면,

$$g \equiv \frac{\partial}{\partial x} = \cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial \phi}, \tag{5}$$

$\partial^2/\partial x^2$ 는 다음과 같이 쓸 수 있다.

$$\begin{aligned} \frac{\partial^2}{\partial x^2} &= \frac{\partial}{\partial x}\cdot\frac{\partial}{\partial x}, \\ &= \frac{\partial g}{\partial x}, \\ &= \frac{\partial r}{\partial x}\frac{\partial g}{\partial r} + \frac{\partial \phi}{\partial x}\frac{\partial g}{\partial \phi}, \qquad (\text{ (2a) 식으로부터}) \\ &= \cos\phi\frac{\partial g}{\partial r} - \frac{\sin\phi}{r}\frac{\partial g}{\partial \phi}, \qquad (\text{(3a) 식과 (3b) 식으로부터}) \\ &= \cos\phi\frac{\partial}{\partial r}\left(\cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial \phi}\right) - \frac{\sin\phi}{r}\frac{\partial}{\partial \phi}\left(\cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial \phi}\right), \\ &= \cos^2\phi\frac{\partial^2}{\partial r^2} + \frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial \phi} - \frac{\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial \phi} \\ &\quad + \frac{\sin^2\phi}{r}\frac{\partial}{\partial r} - \frac{\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial \phi\partial r} + \frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial \phi} + \frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial \phi^2}. \\ &= \boxed{\cos^2\phi\frac{\partial^2}{\partial r^2} + \frac{2\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial \phi} - \frac{2\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial \phi} + \frac{\sin^2\phi}{r}\frac{\partial}{\partial r} + \frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial \phi^2}.} \end{aligned} \tag{6}$$

같은 방식으로 h 를 다음과 같이 정의하면

$$h \equiv \frac{\partial}{\partial y} = \sin\phi \frac{\partial}{\partial r} + \frac{\cos\phi}{r}\frac{\partial}{\partial \phi}, \tag{7}$$

$\partial^2/\partial y^2$ 는 다음과 같이 쓸 수 있다.

$$\begin{aligned}
\frac{\partial^2}{\partial y^2} &= \frac{\partial}{\partial y}\cdot\frac{\partial}{\partial y}, \\
&= \frac{\partial h}{\partial y}, \\
&= \frac{\partial r}{\partial y}\frac{\partial h}{\partial r} + \frac{\partial \phi}{\partial y}\frac{\partial h}{\partial \phi}, \qquad \text{(Eq. (2b) 식으로부터)} \\
&= \sin\phi\frac{\partial h}{\partial r} + \frac{\cos\phi}{r}\frac{\partial h}{\partial \phi}, \qquad \text{((3a) 식과 (3b) 식으로부터)} \\
&= \sin\phi\frac{\partial}{\partial r}\left(\sin\phi\frac{\partial}{\partial r} + \frac{\cos\phi}{r}\frac{\partial}{\partial\phi}\right) + \frac{\cos\phi}{r}\frac{\partial}{\partial\phi}\left(\sin\phi\frac{\partial}{\partial r} + \frac{\cos\phi}{r}\frac{\partial}{\partial\phi}\right), \\
&= \sin^2\phi\frac{\partial^2}{\partial r^2} - \frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial\phi} + \frac{\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial\phi} \\
&\quad + \frac{\cos^2\phi}{r}\frac{\partial}{\partial r} + \frac{\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial\phi\partial r} - \frac{\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial\phi} + \frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}. \\
&= \boxed{\sin^2\phi\frac{\partial^2}{\partial r^2} - \frac{2\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial\phi} + \frac{2\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial\phi} + \frac{\cos^2\phi}{r}\frac{\partial}{\partial r} + \frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}.}
\end{aligned} \tag{8}$$

(6) 식과 (8) 식으로부터, 다음과 같은 결과를 얻는다.

$$\begin{aligned}
\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} &= \cos^2\phi\frac{\partial^2}{\partial r^2} + \frac{2\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial\phi} - \frac{2\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial\phi} + \frac{\sin^2\phi}{r}\frac{\partial}{\partial r} + \frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2} \\
&\quad + \sin^2\phi\frac{\partial^2}{\partial r^2} - \frac{2\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial\phi} + \frac{2\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial\phi} + \frac{\cos^2\phi}{r}\frac{\partial}{\partial r} + \frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}, \\
&= (\sin^2\phi + \cos^2\phi)\frac{\partial^2}{\partial r^2} + \frac{(\sin^2\phi + \cos^2\phi)}{r}\frac{\partial}{\partial r} + \frac{(\sin^2\phi + \cos^2\phi)}{r^2}\frac{\partial^2}{\partial\phi^2}, \\
&= \boxed{\frac{\partial^2}{\partial r^2} + \frac{\partial}{r\partial r} + \frac{\partial^2}{r^2\partial\phi^2}.}
\end{aligned} \tag{9}$$

2. **(총 20 점)**

$$\mathbf{R} = (X, Y) = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}, \tag{10}$$

$$\mathbf{r} = (x, y) = \mathbf{r}_2 - \mathbf{r}_1, \tag{11}$$

위 식에서 $\mathbf{r}_1 = (x_1, y_1)$ 이고 $\mathbf{r}_2 = (x_2, y_2)$ 이다.

(a) (10) 식과 (11) 식으로부터 X 와 x 에 대해 다음과 같이 나타낼 수 있고,

$$X = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}, \tag{12}$$

$$x = x_2 - x_1, \tag{13}$$

일차 편도함수는 다음과 같이 얻어진다.

$$\left(\frac{\partial X}{\partial x_1}\right)_{x_2} = \frac{m_1}{m_1+m_2}, \tag{14}$$

$$\left(\frac{\partial x}{\partial x_1}\right)_{x_2} = -1, \tag{15}$$

$$\left(\frac{\partial X}{\partial x_2}\right)_{x_1} = \frac{m_2}{m_2+m_2}, \tag{16}$$

$$\left(\frac{\partial x}{\partial x_2}\right)_{x_1} = 1. \tag{17}$$

연쇄법칙을 이용하면 다음과 같은 결과를 얻는다.

$$\begin{aligned} \frac{\partial}{\partial x_1} &= \frac{\partial X}{\partial x_1}\frac{\partial}{\partial X} + \frac{\partial x}{\partial x_1}\frac{\partial}{\partial x}, & (18)\\ &= \frac{m_1}{m_1+m_2}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}, & (19) \end{aligned}$$

그리고,

$$\begin{aligned} \frac{\partial}{\partial x_2} &= \frac{\partial X}{\partial x_2}\frac{\partial}{\partial X} + \frac{\partial x}{\partial x_2}\frac{\partial}{\partial x}, & (20)\\ &= \frac{m_2}{m_2+m_2}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}. & (21) \end{aligned}$$

(b) 같은 방법으로, Y 와 y 에 대해 다음과 같이 나타낼 수 있고,

$$Y = \frac{m_1 y_1 + m_2 y_2}{m_1+m_2}, \tag{22}$$

$$y = y_2 - y_1, \tag{23}$$

일차 편도함수는 다음과 같이 얻어진다.

$$\left(\frac{\partial Y}{\partial y_1}\right)_{y_2} = \frac{m_1}{m_1+m_2}, \tag{24}$$

$$\left(\frac{\partial y}{\partial y_1}\right)_{y_2} = -1, \tag{25}$$

$$\left(\frac{\partial Y}{\partial y_2}\right)_{y_1} = \frac{m_2}{m_2+m_2}, \tag{26}$$

$$\left(\frac{\partial y}{\partial y_2}\right)_{y_1} = 1. \tag{27}$$

연쇄법칙을 이용하면 다음과 같은 결과를 얻는다.

$$\begin{aligned} \frac{\partial}{\partial y_1} &= \frac{\partial Y}{\partial y_1}\frac{\partial}{\partial Y} + \frac{\partial y}{\partial y_1}\frac{\partial}{\partial y}, & \\ &= \frac{m_1}{m_1+m_2}\frac{\partial}{\partial Y} - \frac{\partial}{\partial y}, & (28) \end{aligned}$$

그리고

$$\begin{aligned} \frac{\partial}{\partial y_2} &= \frac{\partial Y}{\partial y_2}\frac{\partial}{\partial Y} + \frac{\partial y}{\partial y_2}\frac{\partial}{\partial y}, & \\ &= \frac{m_2}{m_2+m_2}\frac{\partial}{\partial Y} + \frac{\partial}{\partial y}. & (29) \end{aligned}$$